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Innovative Product Development

Title: Learnova — AI-powered redesign of outdated lecture slides into illustrated, interactive teaching decks

Domain: Education Technology · Artificial Intelligence · Document Understanding

Group No. 1 · Department of Artificial Intelligence and Data Science, DJSCE · Interdisciplinary Project Review, Odd Sem 2026

IDEA & PROPOSED SOLUTION

From text walls to teaching decks — without losing a single point the teacher wrote

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IPD Review · AI & DS, DJSCE

In five real course decks (306 slides) we found 83 untitled slides, equations pasted as pictures, and not one diagram, quiz or interactive element

Proposed solution: Faculty reuse the same slide decks for years, and students revise from them before every examination. Learnova is a web platform that takes the deck a teacher already has — a PowerPoint, a slide PDF, Word notes or a typed topic — and rebuilds it as an **editable PowerPoint, an interactive web deck and a study pack**. Every point of the original is kept; what changes is how it is shown: a visual for every idea, an illustration on every slide, and quizzes, solved problems and labs that let students work with the material instead of only reading it.

Detailed explanation of the solution

- **Slide reader:** PDF/PPTX read page by page — titles, bullet levels, tables, figures; OCR for picture-only slides.
- **Decision engine:** 27 detectors choose process, timeline, comparison, definition, chart or cards — with a reason.
- **Assertion titles:** every title states the slide's point (AI with key, rules without).
- **Illustrations:** AI-written diagram, teacher's figure or offline scene on every slide, each with an HTML link.
- **Interaction:** checkpoint quizzes, exact step-by-step solvers, labs for NLP, ML and IPCV topics.
- **Outputs:** editable PPTX, Reveal.js web deck, flashcards, Anki, quiz bank, share link, tutor.

How it addresses the problem

- **Text walls →** one idea per slide, drawn as a visual at a readable density.
- **Untitled slides →** 83 → 0; split parts are named after what they cover.
- **Pictures of equations →** kept as full-size figures; never replaced by OCR noise.
- **Passive reading →** quizzes, steppers and labs on the slide itself.
- **Hours of redesign →** a finished deck in about 10–60 s, fully editable.
- **Unreliable AI →** source text kept verbatim; 98–100% of facts retained.

Innovation and uniqueness

- **Redesign, not generate:** starts from the teacher's deck; Gamma, Copilot and Canva start from a prompt.
- **Faithful by construction:** every AI output is checked against the slide; numbers are computed.
- **Explainable design:** each visual choice comes with its rule and confidence.
- **Glyph-measured layout:** no overflow and no word ever broken across lines.
- **Two fallbacks:** 5-key Gemini pool → Groq/NVIDIA → full offline logic.
- **Built-in labs:** Region Growing, Split & Merge, CYK, Viterbi, NB-WSD, HMM tagging.

PHASE 1 · Sep–Dec 2025

Research & design
Literature survey, competitor study, problem statement, architecture, UI plan

PHASE 2 · Jan–Mar 2026

Prototype V1
Parsing, AI rewriting, engagement score, quiz generation, Streamlit demo

PHASE 3 · Apr–Jul 2026

Learnova V2 platform
FastAPI + React, decision engine, PPTX & web builders, OTP accounts, library

PHASE 4 · Aug–Oct 2026

Real lecture decks
Slide reader + OCR, illustrations, explainers, labs, AI keys, papers

LIVE CODE
Full source, 755 tests, papers and demo decks on GitHub
SDP42/LearnovaV2

LITERATURE REVIEW / EXISTING SYSTEMS

What research and today's tools do — and the gap Learnova fills

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#	Reference / existing system	Method & key findings	Gap / scope for improvement
1	Hu & Wan, "PPSGen", IEEE TKDE 27(4), 2015	Learns sentence importance with regression; ILP selects sentences into slides for academic papers.	Papers only; summarises so content is lost; no visuals; cannot redesign existing slides.
2	Fu et al., "DOC2PPT", AAAI 36, 2022	Hierarchical neural summariser + figure selection + layout prediction from paper–slide pairs.	Needs training pairs; abstractive text can alter facts; input must be a scientific paper.
3	Sun et al., "D2S", NAACL 2021	Slide titles used as queries; retrieval + QA-style summarisation fills each slide.	Text only; no figures, quizzes or interaction; not for an existing lecture deck.
4	Maynez et al., "Faithfulness in abstractive summarization", ACL 2020	Abstractive models hallucinate and omit content in a large share of outputs.	Motivates keeping source text verbatim and checking every AI output.
5	Garner & Alley, assertion–evidence, IJEE 29(6), 2013; Mayer, Multimedia Learning, 2009	Sentence headlines + visual evidence beat bullet lists; one idea per slide, words near pictures.	Design guidance only — no tool applies it automatically to old decks.
6	Sweller & Cooper 1985; Hundhausen et al. 2002 (worked examples, algorithm visualisation)	Studying solved steps and interacting with a visual simulation improves learning.	Lecture slides show finished answers; no step-by-step or interactive layer.
7	Gamma, MS Copilot in PowerPoint, Gemini in Slides, Canva Magic Design (2026)	LLM writes an outline, then fills a template; attractive new decks from a prompt.	Rewrite/merge the teacher's points, drop figures, no labs/solvers, need internet & paid plans.

Gap → how Learnova closes it

Generates new decks	Redesigns the existing deck
Paraphrases, drops items	Verbatim content + fact check
Loses figures/equations	Keeps every figure; OCR only when reliable
No visuals by meaning	27-detector decision engine
No illustration	AI diagram / own figure / offline scene
No interaction	Quizzes, solvers, labs
Cloud & paid only	Offline mode, free keys optional

Common limitations found: research systems summarise scientific papers rather than lecture decks, and they trade faithfulness for brevity; commercial tools produce attractive slides but cannot be checked against the source, ignore picture-only and equation slides, add no interaction and require a connection and a subscription. None of them redesigns the deck a teacher already uses — which is exactly the gap Learnova targets.

TECHNICAL APPROACH — ARCHITECTURE

Six layers, each replaceable; every AI step has an offline fallback

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1 User & client layer

- **Web app** — React · Vite · Tailwind
- **Web deck** — Reveal.js, click reveals
- **PowerPoint** — editable native shapes
- **Mobile PWA** — installable, offline
- **Share link** — view-only, no login

2 API & service layer (FastAPI)

- **Auth** — phone OTP, tokens, rate limit
- **Jobs** — upload · generate · progress
- **Deck library** — versions · refresh
- **Explain / labs / art** — public pages
- **Tutor** — answers from the deck

3 Processing pipeline

- **Reader + OCR** — PDF/PPTX/DOCX, Tesseract
- **Planner** — split · merge · agenda · quiz
- **Decision engine** — 27 detectors, reasons
- **Layout engine** — glyph-measured fitting
- **Audit** — overflow · breaks · facts

4 AI layer (optional)

- **LLM router** — Gemini ×5 → Groq → NVIDIA
- **AI polish** — titles · explanations
- **Topic diagrams** — validated SVG
- **AI labs** — Node-checked JS
- **Offline rules** — always available

5 Output & data layer

- **PPTX builder** — python-pptx
- **Web builder** — self-contained HTML
- **Study pack** — cards · Anki · quiz bank
- **SQLite + files** — users · jobs · decks
- **Caches** — LLM · art · OCR · labs

Technologies used

- **Front end** React 18, Vite, Tailwind CSS, Reveal.js, PWA (manifest + service worker)
- **Back end** Python 3.12, FastAPI, Pydantic, background job queue, SQLite
- **Documents** python-pptx, PyMuPDF, LibreOffice (headless PPTX→PDF), python-docx, Tesseract OCR
- **AI / NLP** Google Gemini (5-key pool, per-key pacing), Groq, NVIDIA NIM via failover router
- **Maths & labs** SymPy exact solvers; JavaScript lab engine; SVG + MuPDF rasterising
- **Quality** pytest — 755 tests; layout audit; gold-set evaluation; e2e API check
- **DevOps** GitHub (SDP42/LearnovaV2), launch configs, deploy: static front end + Docker back end

Data flow

Input (deck / notes / topic) → Processing (read → plan → decide → illustrate → build → audit) → Output (PPTX, web deck, study pack, share link). Jobs run in the background with live stage progress; results are cached by content hash.

TECHNICAL APPROACH — METHODOLOGY

Nine stages from upload to audited deck, with the algorithm behind each

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1 Convert
PPTX → PDF (LibreOffice), DOCX/text → Markdown IR; colour palette of the source kept.

2 Read slides
Furniture = text on ≥ max(3, 0.3n) pages; bullet level $\ell(x) = \text{argmin}\ |x - a_k|$; glued titles split by DP.

3 OCR decision
$\gamma = |\text{ink} \setminus \text{text}| \ / \ |\text{ink}|$; OCR used only if conf ≥ 62, $\gamma \le 0.42$, < 15% symbol lines.

4 Plan
Split overloaded slides at logical boundaries; merge thin parts; agenda, glossary, quizzes.

5 Decide visual
27 detectors → (family, confidence, data, reason); $d^* = \text{argmax}\ c$ after exclusion rules.

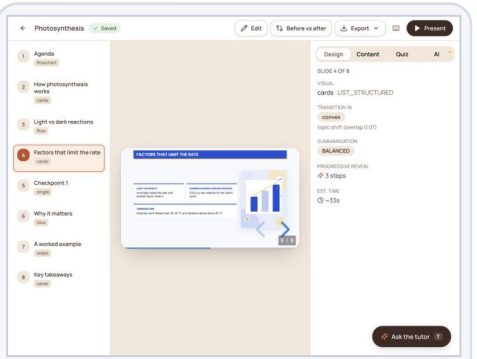
6 AI polish
Assertion titles, explanations, key terms — accepted only if grounded in the slide's words.

7 Illustrate
LLM SVG → whitelist + ≥ 6 shapes + render check → cache; else own figure / offline scene.

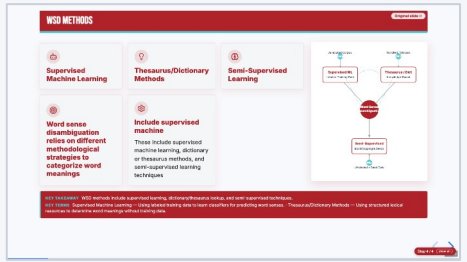
8 Lay out & build
s^* = largest size that fits (≥ 11 pt); $\sigma = \min(1, W / w(\text{longest word}))$; PPTX + web deck.

9 Audit
Overflow, broken words, sparse slides; fact retention $R = |F_{src} \cap F_{deck}| \ / \ |F_{src}|$.

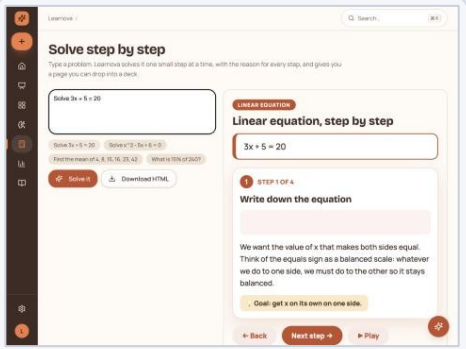
WORKING PROTOTYPE · real screens from the running application



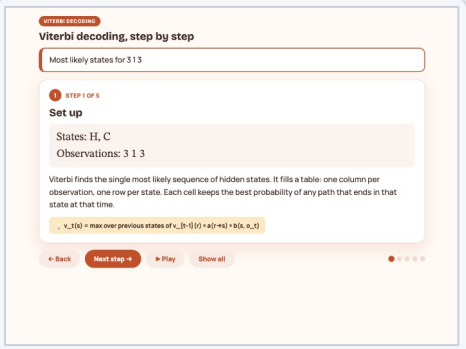
Deck preview — visual chosen and why



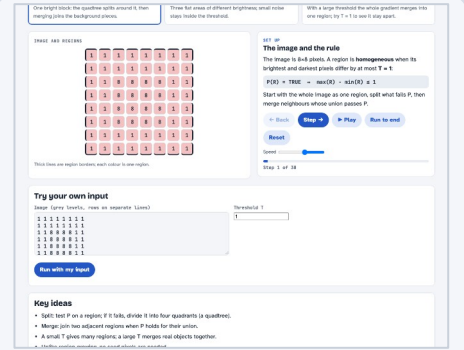
Web deck — AI diagram beside the content



Solve step by step



Viterbi explainer, exact



Split & Merge lab (IPCV)

BEFORE vs AFTER

The problems teachers and students faced — and what Learnova now achieves

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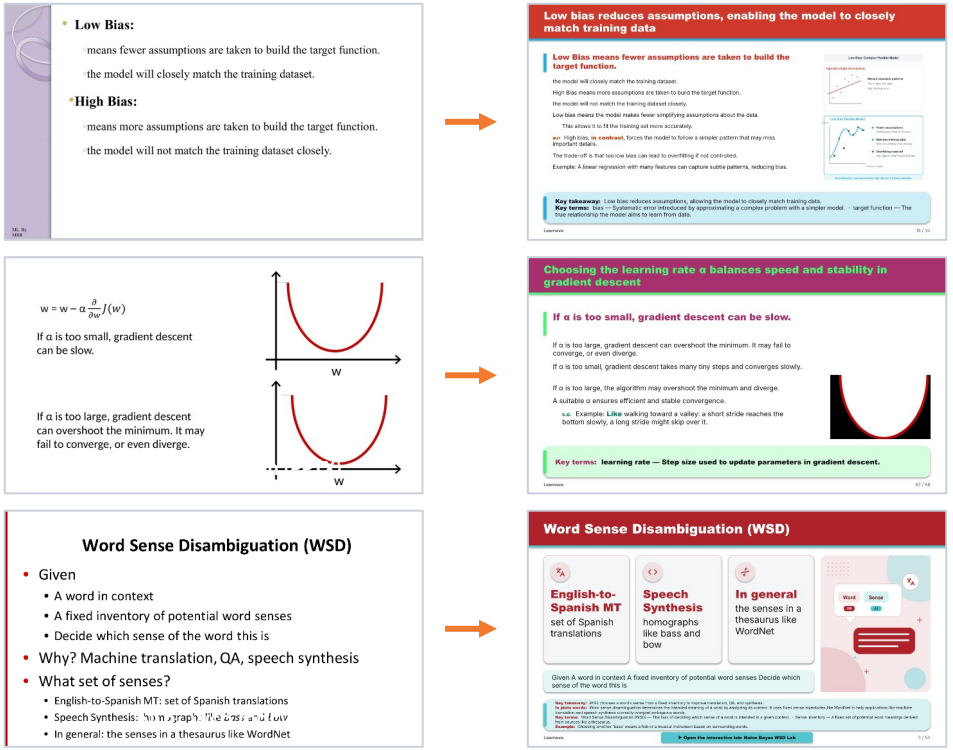
Area	BEFORE — problem faced	AFTER — with Learnova
Text-heavy slides	✘ 100+ words on a slide; students skim and miss points	✔ One idea per slide, drawn as a visual at a readable density
Pictures of equations	✘ Content locked in images; tools drop these slides	✔ Kept full size as figures; OCR only when reliable
Untitled / generic titles	✘ “Slide 9”, “Hypothesis (2/5)” says nothing	✔ Assertion titles; 83 untitled slides → 0
No diagrams	✘ Plain text with nothing to explain the idea	✔ Illustration on every topic slide + HTML link
No interaction	✘ Solved problems shown as finished tables	✔ Quizzes, step-by-step solvers, labs
AI changes content	✘ Paraphrased, merged, invented facts	✔ Verbatim + checked: 98–100% facts kept
Redesign takes hours	✘ An evening per lecture, needs design skill	✔ ≈10–60 s per deck, fully editable
Needs internet / paid AI	✘ Cloud-only, subscriptions, quotas	✔ Offline mode; 5 free keys rotate

Reading the comparison

The left column lists the problems we saw in our own courses: decks that are hard to read, half their content locked in pictures and no way for a student to check understanding. The right column is what Learnova does to the same material, measured on five real decks (two machine-learning PowerPoints and three NLP PDFs). The slide pairs show the same source slide before and after; the counters on the right sum the changes across all five decks.

The content itself does not change: fact retention stayed between 98% and 100% for every deck, and the layout audit found no overflowing text and no broken words.

ORIGINAL → LEARNOVA



83 → 0
untitled slides

251
slides drawn as visuals

190
linked illustrations

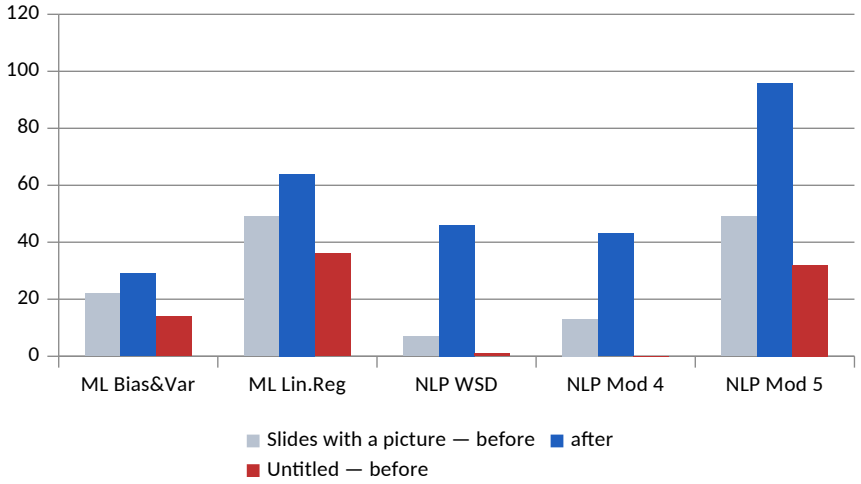
64
quizzes + labs added

RESULTS, IMPACT & BENEFITS

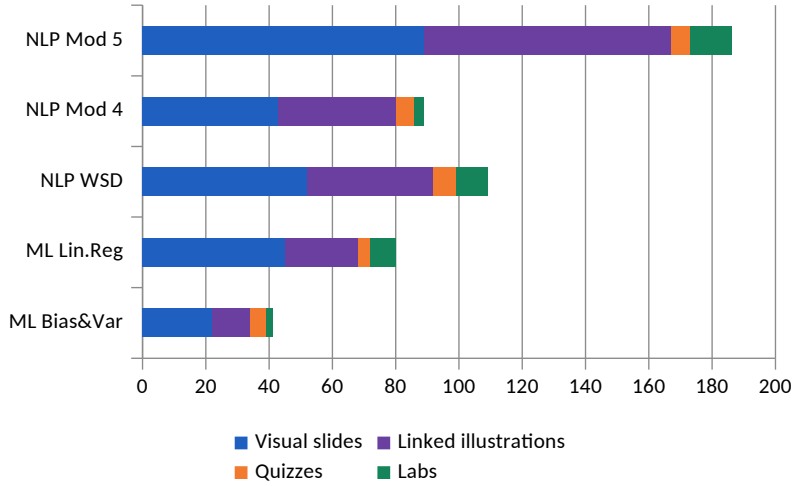
Measured on real course material — not on promises

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Pictures and titles per deck



Added by Learnova (all were 0 before)



97%
visual chosen correctly
101 gold slides, 3 sets

98–100%
facts kept
7 real decks

0
broken words / overflow
layout audit

26/26
problems solved right
held-out benchmark

< 1 ms
decision per slide
rule engine

~35 s
216-page PDF → deck
no API key

Students

- ✔ One clear idea per slide with a picture
- ✔ Steps with reasons for every solved problem
- ✔ Quizzes, labs and flashcards to practise
- ✔ Revise on a phone, even offline

Faculty

- ✔ Old decks modernised in about a minute
- ✔ Their own words and figures kept
- ✔ Editable PowerPoint, ready to teach
- ✔ Quizzes and labs with no extra work

Institution

- ✔ Consistent, accessible course material
- ✔ No paid AI licence needed
- ✔ Supports flipped and remote classes
- ✔ Reusable lab library across subjects

Society · SDG 4

- ✔ Quality education with low-cost tools
- ✔ Works where internet is unreliable
- ✔ Plain-language help for first-generation learners
- ✔ Open formats, easy to share

Use cases: a teacher uploads last year's PPT the night before a lecture and presents the redesigned deck; a student opens the shared web deck to revise with quizzes and labs; an IPCV class explores region growing and split-and-merge in the browser; a college without paid AI tools uses the offline mode. Next: a classroom study of learning outcomes with our own batch.

FEASIBILITY & VIABILITY

Running today on a laptop · pilot-ready for a department · every risk has a named safeguard

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TECHNICAL
Already working end to end: FastAPI back end, React front end, 755 automated tests, layout audit. Each stage is a separate module, so features are added without rework.

DATA
Input is the college's own lecture material (PPTX, PDF, DOCX). Tested on seven real decks of 15 to 216 slides from NLP, ML and IPCV.

ECONOMIC
Free to run offline; Gemini free-tier keys (pool of 5) are optional. Hosting a department server costs less than a single paid AI-presentation licence per year.

OPERATIONAL
Teachers keep control: every slide stays editable and every automatic choice shows its reason. Students need only a browser.

SECURITY
OTP sign-in, hashed codes, rate limits; AI-written SVG and lab code are whitelisted and statically checked; share links are view-only.

Challenges / risks → our safeguard

1. **Messy lecture PDFs**
   symbol bullets, glued titles, picture slides → page reader, DP title split, OCR decision
2. **AI hallucination**
   paraphrase, invented facts → grounded checks, verbatim source, SymPy numbers
3. **API quota / cost**
   free-tier limits → 5-key pool, per-key pacing, failover, offline mode, cache
4. **Broken layout**
   overflow, split words → glyph-measured fitting, 11 pt floor, continuation slides
5. **Slow large decks**
   216 pages, many diagrams → parallel generation, content-hash caching
6. **Unsafe generated code**
   scripts in SVG/labs → element whitelist, node --check, no network APIs
7. **Wrong visual choice**
   rule misfires → alternatives shown in the editor; slide stays editable

Adoption path

1 · Pilot
AI & DS department: our own courses this semester, feedback from 3–4 faculty

2 · Department
shared server, deck library per teacher, labs library grows with faculty labs

3 · College / LMS
Moodle / Google Classroom export, regional languages, analytics

Deployment plan

- **Front end:** static React build on Vercel / Netlify.
- **Back end:** Docker container (FastAPI + LibreOffice + Tesseract) on Render, Railway or Google Cloud Run with a persistent volume.
- **Config:** API URL for the front end; LEARNOVA_PUBLIC_URL so PowerPoint lab and illustration links resolve.
- **Why not all on Vercel:** long jobs, system binaries and on-disk storage do not fit serverless functions.

RESEARCH & REFERENCES

Papers we built on · tools we compared · our own publications and code

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SLIDE GENERATION

- Hu & Wan, PPSGen, IEEE TKDE 27(4):1085–1097, 2015
- Fu et al., DOC2PPT, AAAI 36:634–642, 2022
- Sun et al., D2S, NAACL 2021, pp. 1405–1418

FAITHFULNESS

- Maynez et al., ACL 2020, pp. 1906–1919
- Ji et al., Survey of hallucination, ACM CSUR 55(12), 2023

LEARNING SCIENCE

- Mayer, Multimedia Learning, CUP, 2009
- Sweller, Cognitive Science 12(2), 1988
- Garner & Alley, IJEE 29(6), 2013

WORKED EXAMPLES & LABS

- Sweller & Cooper, Cogn. & Instr. 2(1), 1985
- Hundhausen et al., JVLC 13(3), 2002
- Wieman et al., PhET, Science 322, 2008

LAYOUT & VISUALS

- Knuth & Plass, Breaking paragraphs into lines, SPE 11(11), 1981
- Mackinlay, APT, ACM TOG 5(2), 1986
- Moritz et al., Draco, IEEE TVCG 25(1), 2019

NLP & OCR

- Jurafsky & Martin, Speech and Language Processing, 3rd ed. draft
- Smith, Tesseract OCR, ICDAR 2007
- Meurer et al., SymPy, PeerJ CS 3:e103, 2017

TOOLS COMPARED

- Gamma (gamma.app)
- Microsoft Copilot in PowerPoint
- Gemini in Google Slides · Canva Magic Design

OUR PAPERS

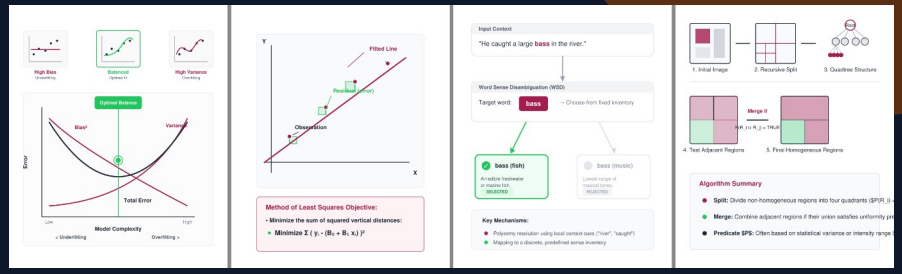
- IEEE conf. paper: Rebuilding lecture slide decks into visually structured, interactive presentations
- Springer LNCS paper: Computed step-by-step explainers and generated labs

Scan to verify: left — full source code, tests and demo decks (github.com/SDP42/LearnovaV2); right — both research papers (IEEE and Springer LNCS), gold sets and the measurement scripts used for every number on these slides. **Web resources:** revealjs.com · python-pptx · PyMuPDF · LibreOffice · Tesseract · ai.google.dev

Thank You

Questions, feedback — and a live demo of Learnova

Upload any lecture deck and watch it become an illustrated, interactive teaching deck.



Diagrams drawn by Learnova for real lecture slides

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Scan for

code, papers & demo decks

github.com/SDP42/LearnovaV2

LEARNOVA · Read · Redesign · Illustrate · Engage · Deliver